Results

Results I

This section presents the exploratory analysis of the shipped dataset. Every figure and the summary table are produced by the EDA analysis orchestrator (`scripts/eda_analysis.py`), which calls the tested figure-data preparers in `src/eda/figures.py`. Running the script regenerates the figures under `output/figures/` and the summary CSV under `output/data/`; the prose below describes what those artifacts show.

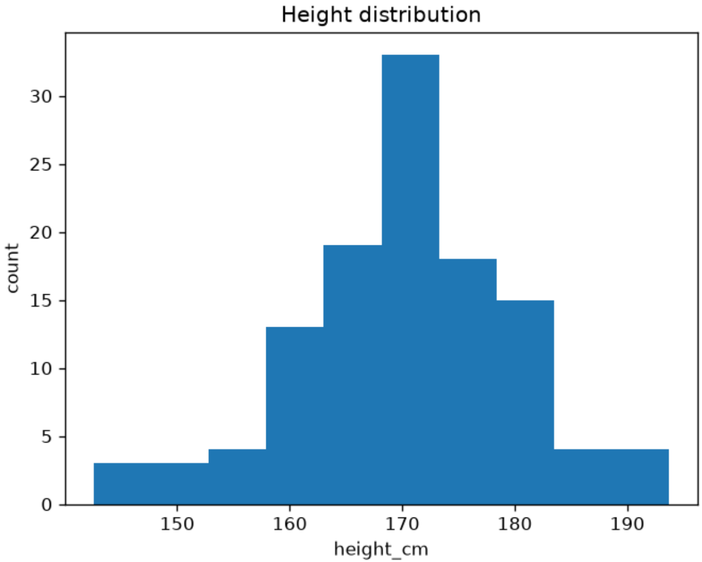
Dataset and missingness I

The dataset contains 120 subject records across three groups (`alpha`, `beta`, `gamma`) with three numeric features: height (cm), weight (kg), and resting heart rate (bpm). A small number of cells are blank by design. `load_dataset()` preserves these as `NaN`; `clean_dataset()` then drops rows missing any numeric feature and reports the count. With the shipped data, four rows are removed, leaving a complete-case dataset for the analysis below.

Distributions I

[@fig:height_histogram] shows the distribution of height across the complete-case dataset, binned by `histogram_data()` and plotted by the analysis script.

Distributions II



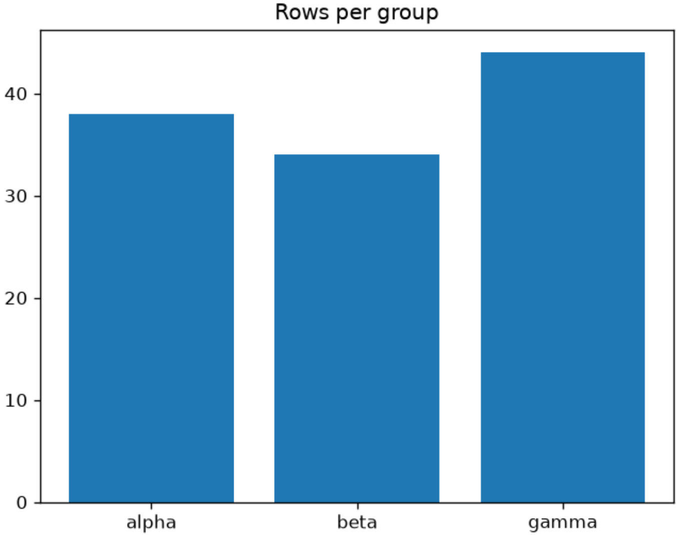
Distributions III

Figure 1: Height distribution: bin counts produced by `histogram_data( frame, "height_cm", bins=10)` in `src/eda/figures.py` and plotted as a bar chart by `scripts/eda_analysis.py`. The bin counts sum to the number of complete-case rows; the shape is the roughly bell-shaped spread expected from the generating process.

Group composition I

[@fig:group_counts] reports how many complete-case rows fall in each group, computed by `group_count_data()` (labels sorted for deterministic output).

Group composition II



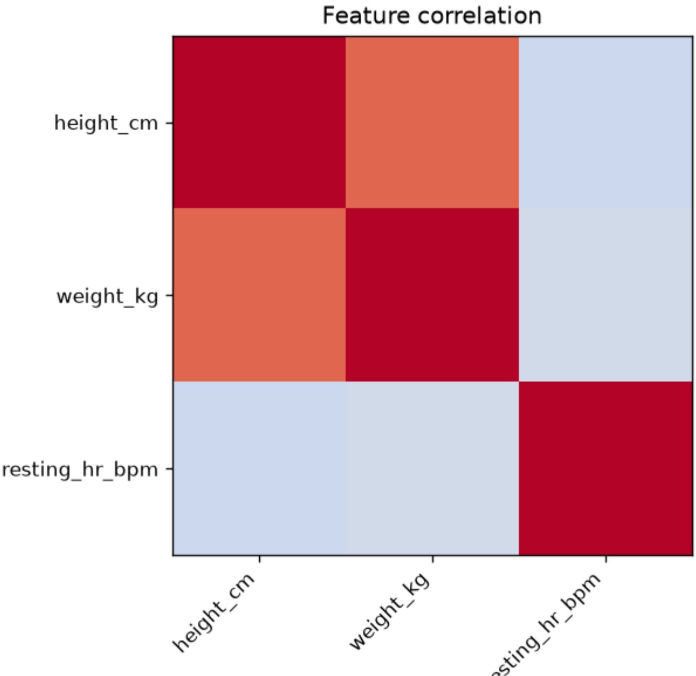
Group composition III

Figure 2: Rows per group: sorted category labels and aligned counts from `group_count_data()`. The three groups are of comparable, though not identical, size; the counts sum to the complete-case row total.

Correlation structure I

[@fig:correlation_heatmap] visualizes the Pearson correlation matrix of the three numeric features, computed by `correlation_matrix()` and prepared for the heatmap by `correlation_heatmap_data()`. The diagonal is unity by construction and the matrix is symmetric.

Correlation structure II



Summary statistics I

The analysis script writes a per-column summary table to `output/data/summary_statistics.csv` from `summary_statistics()`. Each row reports the count of non-missing observations, mean, sample standard deviation, minimum, median, and maximum for one numeric feature.

Table 1: Summary-statistics table written by the analysis script from `src/eda/statistics.py::summary_statistics()`. The concrete numbers are reproduced verbatim by running the script — the manuscript intentionally does not transcribe volatile values, so prose and CSV cannot drift.

Column	Reported statistics
`height_cm`	count, mean, std, min, median, max
`weight_kg`	count, mean, std, min, median, max
`resting_hr_bpm`	count, mean, std, min, median, max

Summary statistics II

`group_means()` complements this with the mean of each numeric feature within each group; the three group means for height are close but not equal, reflecting the mild group structure in the generating process.

Validation I

The analysis was validated through the zero-mock `tests/` suite:

- **Library tests** assert exact statistics, correlation signs, and bin counts against the shipped CSV and hand-built frames.
- **Script test** runs `run_eda()` against a temporary output root and confirms real PNG figures and a real summary CSV are written.
- **Notebook test** confirms the walkthrough notebook binds to the library's public surface and carries no logic in its cells.

All tests pass with coverage exceeding the 90% project gate, with no mocks.

Discussion I

The results confirm the EDA workflow end to end: missingness is surfaced and removed with an explicit count, distributions and group composition are read straight from tested preparers, and the correlation ranking recovers the designed height–weight relationship. The same functions back the interactive notebook, the headless script, and this manuscript — which is the architectural point of the exemplar. Because every number is produced by a tested function and regenerated on demand, the prose here describes structure and provenance rather than transcribing values that would drift the moment the data changed.